X-LineNet: Detecting Aircraft in Remote Sensing Images by a Pair of Intersecting Line Segments

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***Abstract*— Motivated by the development of deep convolution neural networks (DCNNs), aircraft detection has gained tremendous progress. State-of-the-art DCNN-based detectors mainly belong to top-down approaches, which enumerate massive potential locations of aircraft with the form of rectangular regions, and then identify whether they are objects or not. Compared with these top-down detectors, this article shows that aircraft detection via a type of bottom-up method can have better performances in the era of deep learning. In this article, we propose a novel bottom-up detector named X-LineNet. It formulates the aircraft detection task as prediction and clustering of paired intersecting line segments inside each target. Aircraft detection is then a purely appearance-based line segments estimation problem, without any rectangular regions classification or implicit features learning. With simple postprocessing, X-LineNet can simultaneously provide multiple representation forms of the detection result: the horizontal bounding box, the oriented bounding box, and the pentagonal mask. The pentagonal mask is a more accurate representation form of aircraft which has less redundancy than that of a rectangular box. Experiments show that X-LineNet outperforms prevalent top-down and region-based detectors on UCAS-AOD, NWPU VHR-10, and DIOR public data sets in the field of aircraft detection.**

***Index Terms*— Aircraft detection, bottom-up methods, deep convolution neural network, remote sensing image.**

I. Introduction

DRIVEN by the rapid development of remote sensing (RS) technologies, RS images with more exceptional resolution and clearer texture can be easily accessed by the modern airborne and spaceborne sensors. In response to the demand for automatic analysis of RS data, object detection has been widely researched, among which the detection of aircraft occupies an important part owning to the successful application in the field of military and air transportation. However, aircraft detection remains a significant challenge on account of the need for accurate and robust detection at present. Moreover, the complex background, the noise, and the variation of the spatial resolution of RS image further increase the difficulty of this task. Conventional methods in aircraft detection mainly depend on manually engineered features. Cheng *et al.* [1] developed a discriminatively trained model to extract histogram of oriented gradient (HOG) [2] feature pyramids from multiple scales. Then, a threshold operation is performed on the response of the model to judge the presence of aircraft. Bai *et al.* [3] selected the structural feature description as the input of ranking support vector machine (SVM) [4] to identify the existence of objects. Besides, Rastegar *et al.* [5] built an aircraft detection system by combining wavelet features with the SVM classifier. Though these previous methods are useful in specific scenes, the generalization capabilities of them need to be further improved in practice.

Inspired by the success of deep convolution neural networks (DCNNs) [6], [7], various methods based on DCNNs have been proposed in the research field of aircraft detection [8]–[10]. These DCNNs-based detectors follow the principle of top-down and region-based models [11]–[13]. Because they can automatically extract features through the backbone networks [14], [15], the accuracy and robustness of them are much improved compared with manually engineered features. Cao *et al.* [10] used selective search methods to generate a broad set of region proposals, and then send them into the classifier as well as modify the final bounding boxes in a regression manner. Similarly, Ding *et al.* [8] adopted a region proposal network (RPN) module [11] to generate the proposals from a large set of anchor boxes [11] that are used to guide the regression task of the network.

Although satisfactory results have been achieved by modern top-down and region-based approaches, they are not without shortcomings. First, they convert aircraft detection into rectangular region classification, but the rectangular region is a lousy proxy for aircraft themselves. Because it includes not only many distracting backgrounds but also conveys little detailed object information, for example, the direction of aircraft. Besides, top-down and region-based detectors rely on

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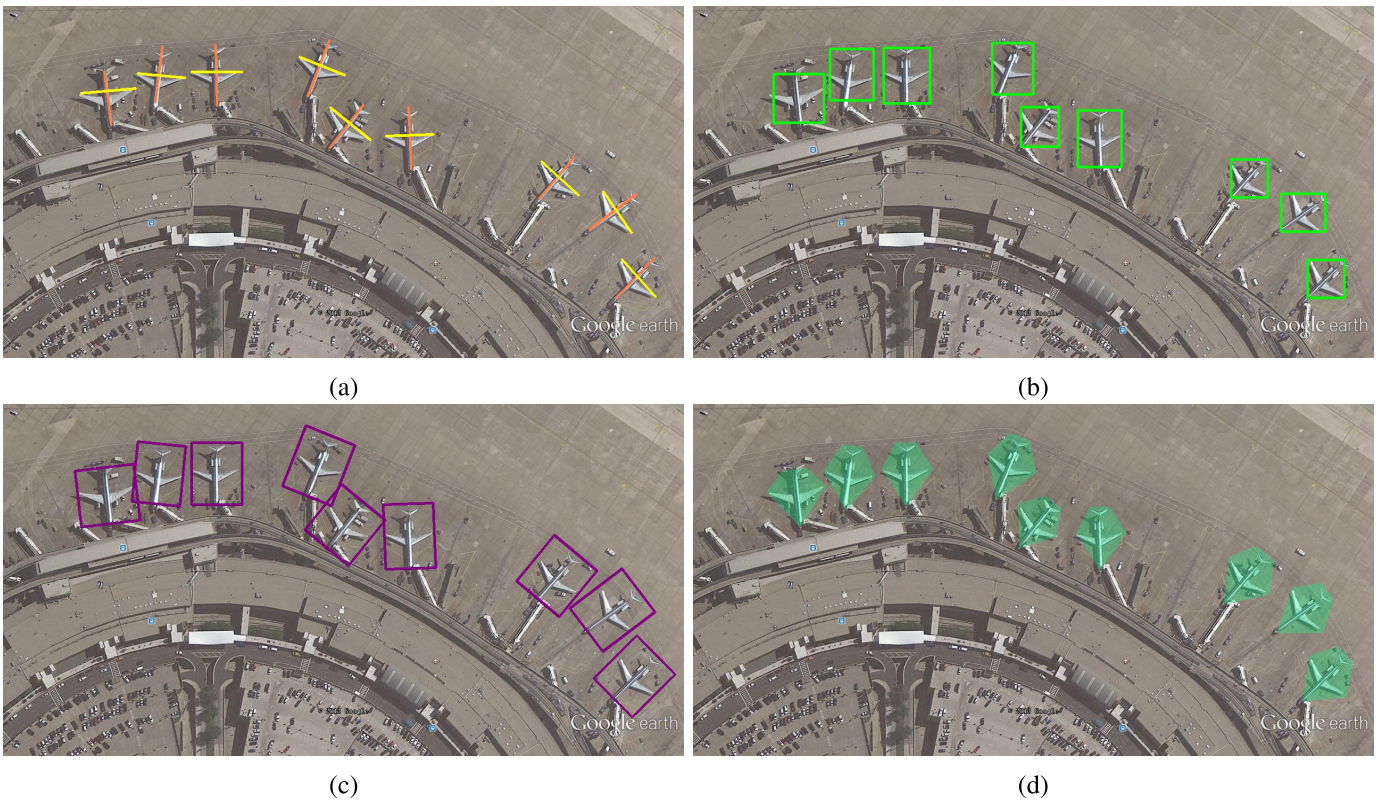


Fig. 1. X-LineNet predicts pairs of line segments to obtain different forms of bounding boxes of aircraft. (a) The line segments predicted by X-LineNet. (b) and (c) The horizontal and oriented bounding boxes generated by the line segments in (a). (d) The pentagonal mask output by X-LineNet.

the anchor mechanism heavily. An extensive set of anchor boxes with different sizes and ratios are densely placed over the different feature maps to cover objects with no omission. Only a small set of anchor boxes have a required level of overlap with the ground truth boxes, which leads to the imbalance in positive and negative samples. This imbalance will hinder the convergence of the network during training. Besides, these detectors regard the anchor boxes with a high ($>$0.7) intersection-over-union (IOU) with any ground truth boxes as regression targets. They do not truly understand the compositional visual grammar information [16]–[18] of the aircraft. Consequently, numerous redundant bounding boxes will be generated, which needs a series of postprocessing, for example, the IOU-based nonmaximum suppression (NMS), to filter the final results. However, the calculations of NMS and IOU are complex and wasteful.

Considering the limits mentioned above in top-down and anchor-based aircraft detectors, we propose a bottom-up and anchor-free model named X-LineNet. X-LineNet is inspired by CornerNet [19] that detects objects via predicting and clustering two corner points. However, CornerNet is not suitable for the task of aircraft detection in RS images. Because a single RS image may contain hundreds of aircraft instances, and this situation will often lead to errors of clustering points detected in CornerNet. Unlike CornerNet, X-LineNet formulates this task as the detection and clustering of paired intersection line segments. These paired line segments are defined as a line from the head of aircraft to tail and a line across two wings. X-LineNet first predicts two heatmaps corresponding to two sets of line segments and then clusters them in pairs with a purely geometry-based approach. As a result of the dimension ascending from point to line, the difficulty of points clustering in CornerNet is solved by X-LineNet. Moreover, the corners in CornerNet often lie outside an object, without stable appearance features. Line segments lie on aircraft; on the other hand, they are visually distinguishable and have rich local appearance features. It is worth noting X-LineNet also abandons the anchor mechanism used in top-down detectors [11], [12], [20], [21] and successfully avoids the problems mentioned above faced by them. Except for the commonly used horizontal and oriented bounding boxes, X-LineNet can provide a new output form of the detection result as the pentagonal mask. It has more detailed semantic descriptions, such as refined position and direction information of each aircraft. Fig. 1 illustrates these output results.

Our contributions and innovations are summarized as follows:

1) We propose a novel aircraft detection model named X-LineNet. It runs in a bottom-up and anchor-free manner. Compared with the dominant top-down and anchor-based methods, X-LineNet shows that the bottom-up approach performs better in this field during the era of deep learning.
2) X-LineNet regards the task of aircraft detection as prediction and clustering of paired appearance-based intersecting line segments. This design enables the network

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to learn rich and specific compositional visual grammar information of aircraft without implicit features learning, for example, the height and width of the corresponding bounding box.

3) Besides horizontal and oriented bounding boxes, one more accurate output as the pentagonal mask can be generated by X-LineNet simultaneously. As a new representation form of aircraft, the pentagonal mask contains less redundant contexts and can provide more semantic descriptions, such as the direction and more accurate localization.

II. Related Works

Object detection has been investigated by a wide variety of methods. In this section, we briefly review the existing deep-learning-based object detection algorithms.

A. Top-Down and Anchor-Based Detectors

In recent years, top-down and anchor-based detectors [8]–[10] have dominated the field of object detection. Generally, they can be classified into two-stage and one-stage detectors.

1) Two-Stage Detectors: Two-stage detectors [11], [15], [22]–[26] decompose the detection task into subtasks of generating and classifying of region proposals. In the early R-CNN [27], region proposals are produced by the algorithm of selective search. It is inefficient because of the generation of numerous redundant proposals. Although SPPNet [28] and Fast R-CNN [29] design particular pooling layers to lift efficiency, both of them are not end-to-end trainable. Later, an RPN [11] is proposed in Faster R-CNN [11] to make the network be end-to-end trainable. The mechanism of RPN can be illustrated as below.

After feature maps extracted by the backbone network [14] from the input image, a large set of anchor boxes will be produced by densely placing rectangular boxes with different sizes and ratios. NMS and scoring mechanisms within the RPN are performed on account of filtering these numerous anchor boxes and obtaining required proposals. Final detection results can be accessed after the classification and modification of these proposals. Benefit from the adoption of RPN [11], Faster R-CNN can be trained end-to-end, and its accuracy further increases.

In the field of object detection in RS images, most state-of-the-art detectors rely on R-CNN frameworks. Cheng *et al.* [30] constructed a rotation-invariant model by introducing a rotation invariance layer into the network and promote the performance of object detection in RS images. Li *et al.* [31] proposed a novel deep-learning-based object detection framework, including RPN and local-contextual feature fusion network, to effectively address the challenges of appearance ambiguity and rotation variations. Xue and Tong [32] proposed a model named DIOD based on advanced RPNs (ARPNs) and weakly semisupervised deep joint sparse learning.

Different from the aforementioned RPN-based detectors, X-LineNet is a purely appearance-based line segments estimation network, without region proposals, and classification.

2) One-Stage Detectors: In pursuit of computation efficiency and inference speed, one-stage detectors [12], [13], [20], [33], [34] are designed without region proposal module. Initially, YOLOv1 [33] without anchor mechanism cannot provide accuracy comparable to that of two-stage detectors. Later, anchor methods are also extensively utilized in one-stage detectors [12], [13], [20], [34] to improve the accuracy, and dense anchor boxes with different scales are placed within specific layers.

Although the gain in the accuracy, there are some problems accompanied by the bringing in anchor mechanism for one-stage detectors. First, a large set of anchor boxes with multi scales are produced to supply sufficient location information, while only a small set of them are positive samples. It leads to the imbalance in positive and negative samples, which will prolong the process of training. Besides, many hyper-parameters are introduced along with anchor boxes, such as concrete settings for sizes and radios of anchor boxes in specific layers, which brings in much inconvenience to adjust them carefully.

On account of the drawbacks of the anchor mechanism mentioned above, there have been some works on modifying it. In RetinaNet [13], focal loss is promoted to optimize the imbalance in positive and negative samples. Furthermore, MetaAnchor [35] is proposed to decrease the reliance of the anchor settings on prior knowledge.

X-LineNet falls into the category of the one-stage detector because the process of region proposal is discarded.

B. Bottom-Up and Anchor-Free Detectors

Considering the above shortcomings of anchor-based detectors, some anchor-free algorithms [16], [19] for natural object detection have emerged. CornerNet [19] and ExtremeNet [16] are two representative detectors of this new field. Their idea is to transform the object detection task into the processes of predicting and clustering keypoints, without region classification. Thereby, they can be seen as detecting objects in a bottom-up method. Although they achieve state-of-the-art results in the MS COCO [36] benchmark, it is not suitable to apply them to aircraft detection in RS images. The reasons are as follows:

For CornerNet [19], the clustering method needs the network to learn some implicit information. Specifically, CornerNet predicts an embedding vector for each detected corner. If a top-left corner and a bottom-right corner belong to the same bounding box, the distance between their embedding vectors should be small. However, this method does not apply to aircraft detection in RS images because hundreds of objects in an image will lead to difficulty in converging the network and embedding errors frequently, as shown in Fig. 2(a).

For ExtremeNet [16], the clustering method is to match four extreme points with the center points of objects. The basic rule is that if the calculated geometric center of four extreme points matches the predicted one, the four extreme points should belong to the same object. This method is still unable to be applied to detect aircraft, because multiple aircraft are symmetrical in many cases, as shown in Fig. 2(b).

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(a) (b) (c)

Fig. 2. (a) and (b) The results of CornerNet and ExtremeNet, respectively; there are some clustering errors in these two figures. (c) The output result of X-LineNet.

Different from CornerNet and ExtremeNet, X-LineNet converts the detection of points to line segments. As a result of the dimension ascending from point to line, difficulties of clustering in CornerNet and ExtremeNet are solved by X-LineNet in the field of aircraft detection. Fig. 2 shows representative contrast results between X-LineNet and the other two bottom-up detectors.

III. X-LineNet

In this section, we first briefly introduce the overall architecture of the proposed X-LineNet. Then, the design of the loss function is elaborated. Next, we introduce the postprocessing processes of the model, including the extraction and clustering of predicted line segments. Third, a novel Cutting-line Algorithm concept utilized to optimize the potential adhesion problem of adjacent line segments is introduced. Finally, we elaborate on how the model outputs the pentagonal mask.

A. Architecture

As shown in Fig. 3, X-LineNet is a simple, unified network composed of a backbone network, and multiple detection heads, depending on different detecting requirements. Each detection head is composed of three cascading prediction modules to predict the corresponding detection target. One prediction module consists of three convolution layers to further extract deep-features. The sizes and channels of their convolution kernels are $3 \times 3 \times 256$, $3 \times 3 \times 256$, and $1 \times 1 \times 1$, respectively. Here, the last channel means the output class of X-LineNet is only one: aircraft. We use two types of activation functions between these convolution layers, among which BN [37] + ReLU is used as the activation function between two 3×3 convolution layers. ReLU is set between 3×3 and 1×1 layers. In the final output layer, we use the sigmoid function to activate it, so that the output values are mapped between 0 and 1.

B. Definition of Loss Function

X-LineNet predicts two heatmaps: one for head–tail line segments and the other for line segments across the wings. Let the shape of the input image be $H \times W$, and the size of each heatmap is $H/D \times (W/D)$, where D corresponds to the output stride of X-LineNet. Each pixel value $y \in [0:1]^{H/D \times (W/D)}$ in heatmaps represents the confidence of being judged as a positive sample (the corresponding line segment).

As shown in Fig. 4(a), let (x_h, y_h) be the head coordinate of the aircraft, and (x_t, y_t) be the tail. The (x_l, y_l) and (x_r, y_r) are the coordinates of two wings, respectively. For the first heatmap, we define segment $L_1[(x_h, y_h), (x_t, y_t)]$ as the regression target. $L_2[(x_l, y_l), (x_r, y_r)]$ is the regression target of heatmap two. During training, as shown in Fig. 4(b), to balance the positive and negative samples to some extent, we map each line segment to an ellipse region, instead of regarding L_1 or L_2 as the ground truth directly. For L_1, parameters of the mapped ellipse are defined as follows:

$$\begin{cases} (x_c, y_c) = \left(\dfrac{x_h + x_t}{2}, \dfrac{y_h + y_t}{2} \right) \\ a = \sqrt{(x_h - x_t)^2 + (y_h - y_t)^2} \\ b = \dfrac{a}{r} \\ \theta = \arctan\left(\dfrac{y_h - y_t}{x_h - x_t} \right) \end{cases} \tag{1}$$

where (x_c, y_c) is the center of the ellipse. a and b represent the lengths of the major and minor axes, respectively. r is a parameter used to control the ratio of lengths of two axes, and θ is the rotation angle of the ellipse. Parameters of the mapped ellipse for L_2 are similar to L_1.

For the ground truth, all pixel values within ellipses are set to 1 as positive samples, and other pixels are regarded as the negative samples with values set to 0. We adopt a pixel-level focal loss [13] as the loss function of X-LineNet. It is defined

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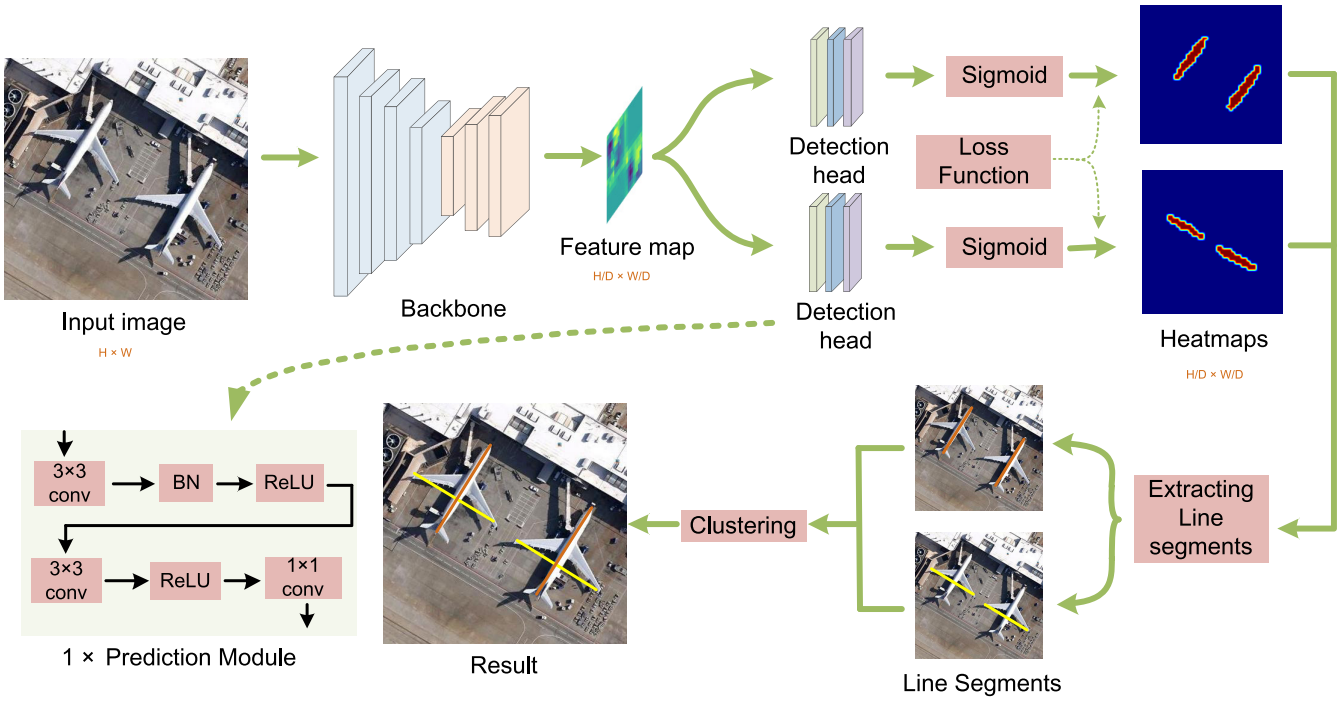


Fig. 3. Architecture of X-LineNet. $H \times W$ is the input image's size, and D is the output stride of the backbone network. "Conv" is the convolution layer, "BN" is the batch normalization layer, and "ReLU" is the ReLU layer.

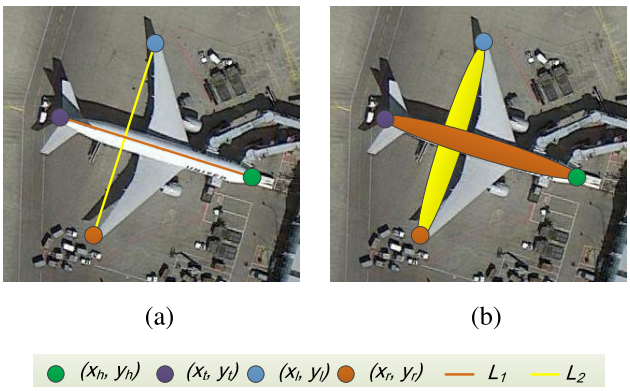


Fig. 4. Definition of paired line segments. The four points in the figure correspond to (x_h, y_h), (x_l, y_l), (x_t, y_t), and (x_r, y_r), respectively. The two segments correspond to L_1 and L_2, respectively.

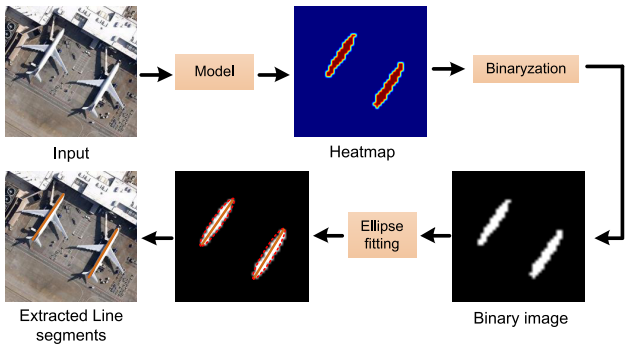


Fig. 5. Extraction of line segments. We transform the heatmap to a binary image and find the fitting ellipse of each line area in the binary image, then, we regard the long axis of an ellipse as the target line segment.

as follows:

$$\mathcal{L}_{oss} = -\frac{1}{N}\sum_{xy}\begin{cases}(1-\acute{V}_{xy})^{\gamma}\log \acute{V}_{xy}, & \text{if } V_{xy}=1\\ \acute{V}_{xy}{}^{\gamma}\log(1-\acute{V}_{xy}), & \text{if } V_{xy}=0\end{cases} \tag{2}$$

where N is the number of aircraft. $\acute{V}_{xy}$ represents the predicted value at the coordinate (x, y) in the corresponding heatmap, and V_{xy} corresponds to the ground truth. γ is a hyper-parameter in original focal loss to adjust the weights of easy and hard samples.

C. Postprocessing

1) Extraction of Line Segments: Each pixel in the predicted heatmap can be classified into: belong to a segment or not. Their values are mapped between 0 and 1 by sigmoid function, representing the probabilities of pixels as positive samples. Thereby, the heatmaps are probability maps, and the extraction of target line segments is a process of pixel-level binary classification at a probability threshold of 0.5. In practice, we use the 0.5 threshold to transform heatmaps to binary images. As shown in Fig. 5, each line segment maps a connected domain in the binary image. We fit each connected domain via an ellipse-fitting algorithm and define each ellipse's long axis as the predicted line segment.

2) Clustering of Paired Line Segments: After the extraction of line segments, the next step is clustering them in pairs. We propose two simple clustering rules according to the geometric relationship of paired line segments. As shown in Fig. 6, the L_1 and L_2 represent the line segments extracted in two heatmaps, respectively. The rules are as follows:

1) L_1 and L_2 intersect.
2) The angle θ between L_1 and L_2 is in the range: $[60° : 120°]$.

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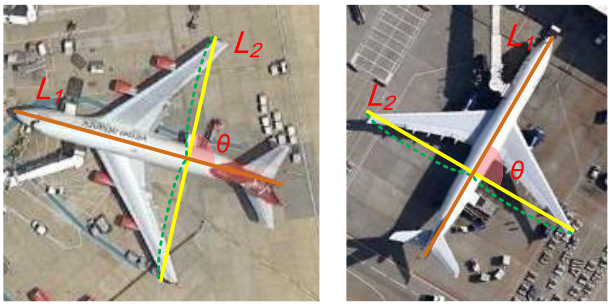


Fig. 6. Illustration of clustering line segments.

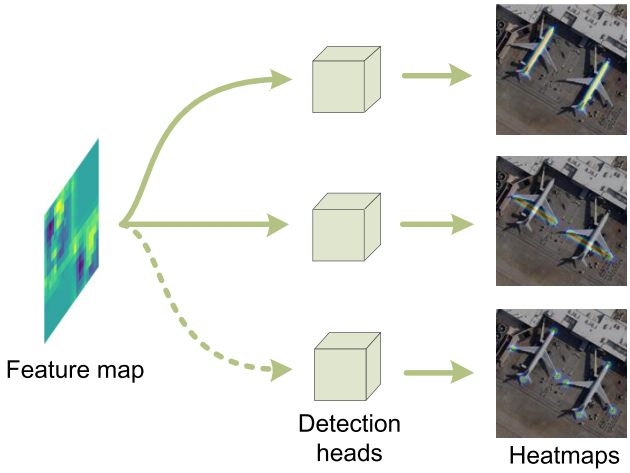


Fig. 7. Extra detection head is added to predict possible adhesion locations, i.e., four endpoints of paired line segments.

D. Cutting-Line Algorithm for Line Segments Adhesion Problem

Because the existence of output stride in X-LineNet, if two aircraft are close to each other, their corresponding line segments maybe stuck together at endpoints. We call this problem as line segment adhesion. To address this problem, we design a Cutting-line Algorithm which acts as separating the target adhesion line segments.

First, the algorithm should inspect adjacent aircraft and locate the positions of adhesion. To achieve this aim, as shown in Fig. 7, we connect another detection head in parallel after the backbone network to predict the possible adhesion locations, that is, four endpoints of paired line segments. This detection head's structure and loss function are the same as that of the line segments, and the only difference is that its regression targets are endpoints instead of lines. For its ground truth, as shown in Fig. 8(a), we map each endpoint to a circular region with a radius of 3. On the one hand, the circular region can balance the positive and negative samples to some extent; on the other hand, it is a guide to locate the adhesion. For the output heatmap, as shown in Fig. 8(b), the sizes of point regions in the positions of adhesion are bigger than other ones, which is the key to the Cutting-line Algorithm to classify each point region into adhesion or not. Like the extraction of line segments, we also extract each point region via the method of finding the connected domain and then introduce

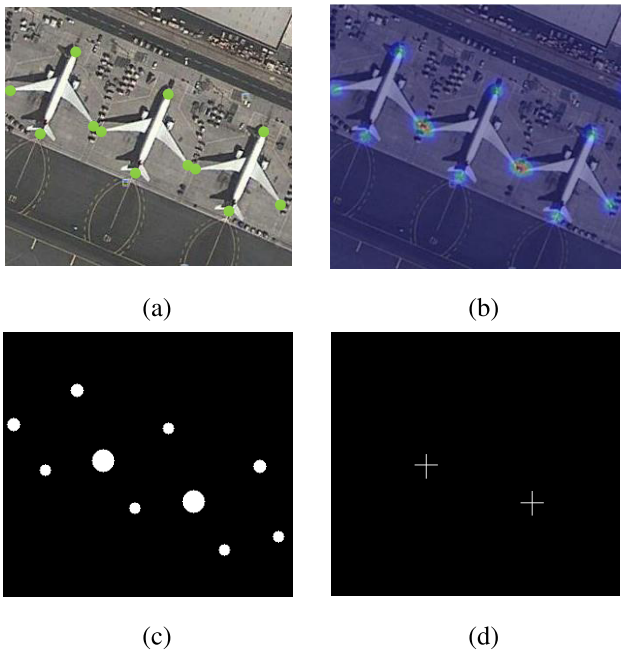


Fig. 8. (a) The ground truth of endpoints detection head and (b) the corresponding predicted heatmap of X-LineNet. (c) The result obtained by the operation of the minimum enclosing circle for the heatmap and (d) the cutting map generated by (c).

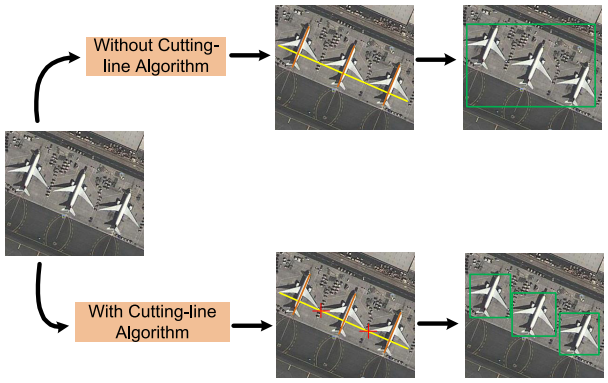


Fig. 9. Effects of Cutting-line Algorithm.

the minimum enclosing circle of each point region to map corresponding predicted endpoint, as shown in Fig. 8(c). Then, we can find out the minimum enclosing circle with the radius exceeds a certain threshold, and there is a high probability that the adhesion problem occurs within it. Here, the threshold is defined as $3n$, 3 is the radius of the ground truth circle, and n is a magnification.

After the adhesion positions being screened, as shown in Fig. 8(d), we can generate a cutting map according to these positions. The cutting map is a binary image, and we design cross-cutting lines within it. The coordinate and length of each cross-cutting line are defined via the corresponding minimum enclosing circle, that is, the center and radius of it. The cutting map can work efficiently by subtracting it from the binary images (mentioned in Section: III-C1) of line segments. Fig. 9 shows the effect of the Cutting-line Algorithm.

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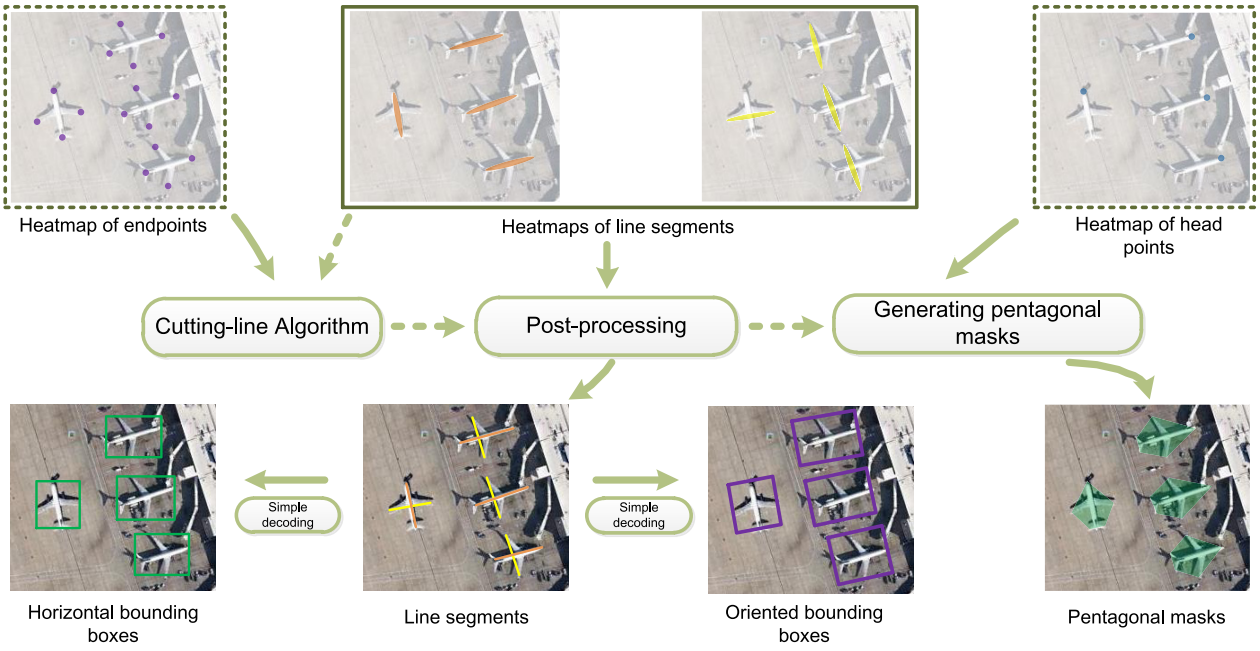


Fig. 10. X-LineNet can output different bounding boxes. Dashed lines represent optional items.

E. Generation of the Pentagonal Mask

As shown in Fig. 10, X-LineNet can output commonly used two result forms (horizontal and oriented bounding boxes) by merely decoding the predicted line segments. In this article, we introduce a better representation form of aircraft. It is the pentagonal mask that has less redundancy than that of the rectangular box.

To generate the pentagonal mask with further direction context of aircraft, X-LineNet needs to identify the specific meanings of these four endpoints of two target line segments. Benefited from the flexibility of the model structure, it can be achieved easily by connecting an extra detection head to predict the head point of aircraft. After extracting each head point, we can match it via Euclidean distance to two endpoints of L_1 (Section: III-B). The endpoint with the most considerable matched value is regarded as the head point, and the other one is the tail. After the head and tail points are found, the two endpoints of L_2 can be identified easily. The corresponding tail point and L_2 generate the base of the pentagonal mask. Its slope and length are defined as the slope of L_2 and one-third of the length of L_2, respectively. X-LineNet can output the pentagonal mask with the above simple operations.

IV. Experiments

In this section, the experiments are implemented to evaluate the detection performance of X-LineNet. We first present the implementations of X-LineNet, such as data sets, training, and test details, backbone networks, and evaluation metric. Then we introduce sufficient ablation experiments. Finally, the comparison with state-of-the-art methods implies the effectiveness of the proposed method.

A. Data Sets

1) UCAS-AOD: There are 7482 aircraft in 1000 large and high-resolution RS images in UCAS-AOD [38] data set.

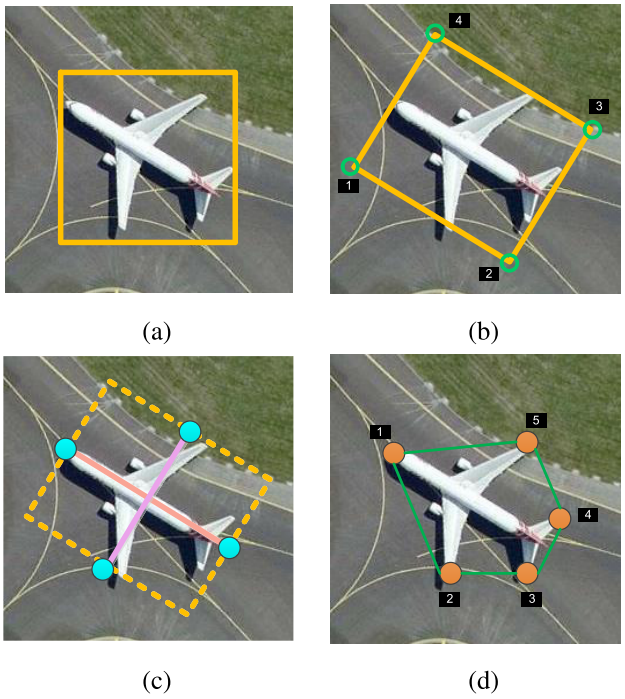


Fig. 11. Annotation formats in UCAS-AOD and Aircraft-KP data sets. The numbers indicate the order of annotation.

As shown in Fig. 11(a) and (b), all aircraft in UCAS-AOD are labeled in two forms: horizontal and oriented bounding boxes. In the training stage, the annotations of line segments used by X-LineNet can be transformed from the oriented bounding box, as shown in Fig. 11(c). Because X-LineNet can output both horizontal and oriented boxes at the same time, we use both two annotation formats in the test stage. We randomly select 800 images used for training, leaving the other 200 for

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testing. To enhance the training set, we take each aircraft as the crop center and divide each large image to crop images into multiscales with 384×384, 512×512, and 800×800. In this way, we can get more than 15 000 aircraft images used to train.

2) Aircraft-KP: In this article, along with X-LineNet, we open source a new aircraft keypoint data set named Aircraft-KP. As shown in Fig. 11(d), each aircraft in this data set is labeled with five keypoints. All images in Aircraft-KP are taken from UCAS-AOD, so the numbers of aircraft and images are equal to the UCAS-AOD. It is worth noting that annotations in Aircraft-KP can be easily transformed into standard horizontal and oriented bounding boxes. Moreover, it has more refined location information than UCAS-AOD, which can further explore the capability of X-LineNet. In the experiments, the settings of training and test sets are the same as UCAS-AOD. Aircraft-KP is available at the URL: https://github.com/Ucas-HaoranWei.

3) NWPU VHR-10: NWPU VHR-10 [30] data set contains about 700 aircraft in 80 images labeled in the form of the horizontal bounding box. To test the generalization performance of X-LineNet, we merge the aircraft data in NWPU VHR-10 to the test set in our experiments.

4) DIOR: DIOR [39] is a large-scale data set of RS images for horizontal object detection. For the aircraft category, there are 705 images with 8212 various aircraft instances in the DIOR test set. We merge all the aircraft data to our test experiments as same as NWPU VHR-10 to further verify the generalization ability of X-LineNet.

B. Backbone Networks

We experiment with three backbone networks: 104-Hourglass [19], ResNet-101 [40], [41], and DLA-34 [41], [42]. Their details are as follows:

1) 104-Hourglass: 104-Hourglass Network [19] is a fully convolutional neural network that consists of two sequential hourglass modules. Each hourglass module is a symmetric five-layer of downsampling and upsampling convolution networks with skip connections. The output stride of 104-hourglass is 4. Although the network is large, it can yield the best line segments prediction performance for X-LineNet.

2) ResNet-101: We use a high-resolution ResNet-101 [40], [41] with deformable convolutional [43] as another backbone. The output stride of this ResNet-101 is 4. Xiao *et al.* [40] augmented a standard residual network [15] with three upsampling convolutional networks to allow for the higher resolution output. To further improve the performance of this ResNet, Zhou *et al.* [41] added one 3×3 deformable convolutional layer before each up-sampling convolution with channels of 256, 128, and 64, respectively.

3) DLA-34: Deep Layer Aggregation with only 34 layers (DLA-34) [42] is a small network that is less than 100M. Multiple hierarchical skip connections guarantee the high-resolution output ($4\times$ output stride) of the network. In our experiments, we use the improved version of DLA-34 modified by Zhou *et al.* [41].

C. Training and Test Details

We train on an input resolution of 512×512, which yields an output resolution of 128×128 for all models with different backbones. For the images in training set with multiscales (384×384, 512×512, and 800×800), they are resized to the uniform size (512×512) by bilinear interpolation, which ensures the diversity of the sizes of aircraft samples. Moreover, we use random rotation and color jittering as data augmentation to get rid of the risk of over-fitting and use Adam [44] to optimize the overall objective. For the ResNet-101 and DLA-34, we train the model with a batch-size of 24 (on 2 GPUs). We set a learning rate of $2.5e-3$ for 60 000 iterations, with a drop of $10\times$ at 40 000 iterations. For 104-hourglass, we use a batch-size of 12 (on 2 GPUs), and other settings are the same as ResNet-101 and DLA-34. We train DLA-34 from scratch, but the 104-Hourglass and ResNet-101 are pretrained [41], [45] on the COCO data set [36] to avoid the risk of over-fitting further. All experiments are performed on two RTX-2080Ti GPUs, one I9-9900K CPU with PyTorch 1.0 [46]. ResNet-101 and DLA-34 train in 10 h, while 104-hourglass requires one day.

We keep the original resolution of each input image during the test stage instead of resizing it to a fixed size. All the test results are obtained from a single scale without bells and whistles.

D. Evaluation Metric

We use average precision (AP) at an IOU threshold of 0.5 following the PASCAL VOC data set [47] to evaluate the performance of X-LineNet. The AP is a mainstream evaluation metric in the field of object detection. Before introducing AP, we first introduce the concepts of TP, FP, FN, Precision, and Recall.

1) TP, FP, and FN: A detection is labeled as true positive (TP) if the IOU between the detection box and a ground truth box is higher than the threshold, otherwise false positive (FP). If a ground truth box has no matching detection boxes, it is labeled as false negative (FN).

2) Precision and Recall: The Precision denotes the fraction of true positives among the detection results, and Recall is the fraction of true positives among the total number of objects. After calculating the number of TP, FP, and FN for the detection result, the Precision and Recall are defined as

$$\text{Precision} = \frac{\text{TP}}{(\text{TP} + \text{FN})} \tag{3}$$

$$\text{Recall} = \frac{\text{TP}}{(\text{TP} + \text{FP})}. \tag{4}$$

3) AP: The P–R curve can be drawn according to the precomputed Precision and Recall of the detection results. The AP metric is calculated by the area under the P–R curve; hence, the higher the AP value, the better the performance, and vice versa. The AP is as follows:

$$\text{AP} = \int_0^1 \text{P}(\text{R}) d(\text{R}) \tag{5}$$

where P and R represent the single-point values of Precision and Recall, respectively.

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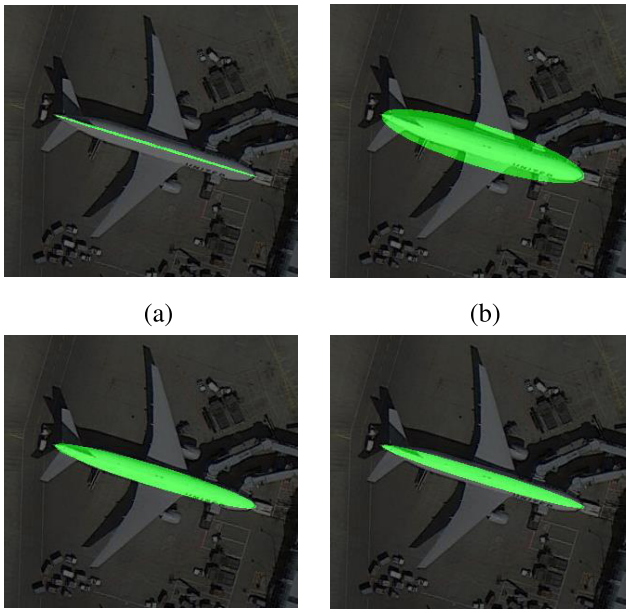


Fig. 12. The ground truth ellipses at different r values; (a), (b), (c), and (d) correspond to different r values with a, 5, 10, and 15, respectively.

TABLE I
COMPARISONS OF DIFFERENT r VALUES

r	a	5	10	15	20
AP	0.925	0.917	0.938	0.935	0.929

E. Ablation Studies

In this section, we conduct ablation experiments to explore the proposed X-LineNet and demonstrate the effectiveness of our method. We use 104-hourglass as the backbone of the model, and utilize the Aircraft-KP data set with the horizontal box to calculate the AP. Qualitative results output by X-LineNet are shown in Fig. 13.

1) Parameters Optimize: As mentioned in Section: III-B, two parameters are involved in the training stage: the ratio r of the lengths of two axes of the ground truth ellipse and the γ in the loss function. First, we set the γ to 2 following the focal loss [13], and test the performance of X-LineNet at different r values with a, 5, 10, 15, 20, respectively. Here, a denotes the length of the ground truth ellipse's major axis, which means the line segment is not mapped. For the structure of X-LineNet, we only keep two basic detection heads to predict paired line segments. Fig. 12 shows different ground truth ellipses with different r values. As shown in Table I, when the value of r is 10, X-LineNet achieves the highest AP of 93.8%. It gives a decent AP improvement of 1.3% compared with the model without ellipse mapping (when r value is a), demonstrating that the design of the ground truth with ellipse mask can balance the positive and negative samples to some extent and improve the performance of X-LineNet.

γ is a parameter of the loss function to adjust the weights of easy and hard samples, and it is set to 2 in the original Focal Loss. To verify whether 2 is also the best value for X-LineNet, we fix the value of parameter r to 10 and experiment with different γ values. As shown in Table II, fortunately, 2 is also the best γ value for our model.

TABLE II
COMPARISONS OF DIFFERENT γ VALUES

γ	0.5	1	2	3	4
AP	0.911	0.919	0.938	0.933	0.927

TABLE III
COMPARISONS OF DIFFERENT n VALUES

n	1	1.5	2	3	∞
AP	0.936	0.948	0.944	0.939	0.937

TABLE IV
COMPARISONS OF DIFFERENT RADIUS

radius	1	2	3	4	5
AP	0.940	0.948	0.948	0.943	0.936

2) Effects of the Cutting-Line Algorithm: To address the potential adhesion problem as mentioned in Section: III-D, we design the Cutting-line Algorithm. There is a threshold $3n$ used to determine whether adhesion occurs in this algorithm. Here, 3 is the radius of the ground truth circle mapped from the corresponding endpoint, and n is a magnification. We test the effects of the Cutting-line Algorithm at different n values with 1, 1.5, 2, 3, and ∞. ∞ means all line segments are nonadhesion by default. The parameters r and γ are set to 10 and 2 based on Tables I and II.

To run Cutting-line Algorithm, we add a detection head to predict four endpoints of paired line segments. As shown in Tables III and II, the AP with the n value of ∞ shows the impact of the third detection head to the basic two is minimal (only 0.1%). The n value of 1.5 yields an AP improvement of 1% compared with the model without the Cutting-line Algorithm (Tables I and II), which proves the algorithm effectively solves the potential line segment adhesion problem.

Conversely, with the n value of 1.5, we test whether 3 is the best radius of the ground truth circle. As shown in Table IV, X-LineNet can achieve the highest 94.8% in AP when the radius is 2 or 3. It is reasonable for us to choose 3 as the radius.

3) Generation of the Pentagonal Mask: In order to output the pentagonal mask, X-LineNet needs four detection heads. We conduct a further experiment to analyze the impact of the fourth detection head. The result shows that X-LineNet with four detection heads can also achieve 94.8% in AP. The generation of the pentagon mask gives no impact on other output forms. For the Aircraft-KP data set, we also test the quality of the pentagonal mask. We take the Mask AP that often used in the field of Instance Segmentation [48] as the evaluation metric. The definition of Mask AP is the same as AP, and the only difference is that the Mask AP uses a pixel-level IOU calculation. The experimental result shows that X-LineNet can achieve 89.3% in Mask AP. The quality of the pentagonal mask output by X-LineNet is very high.

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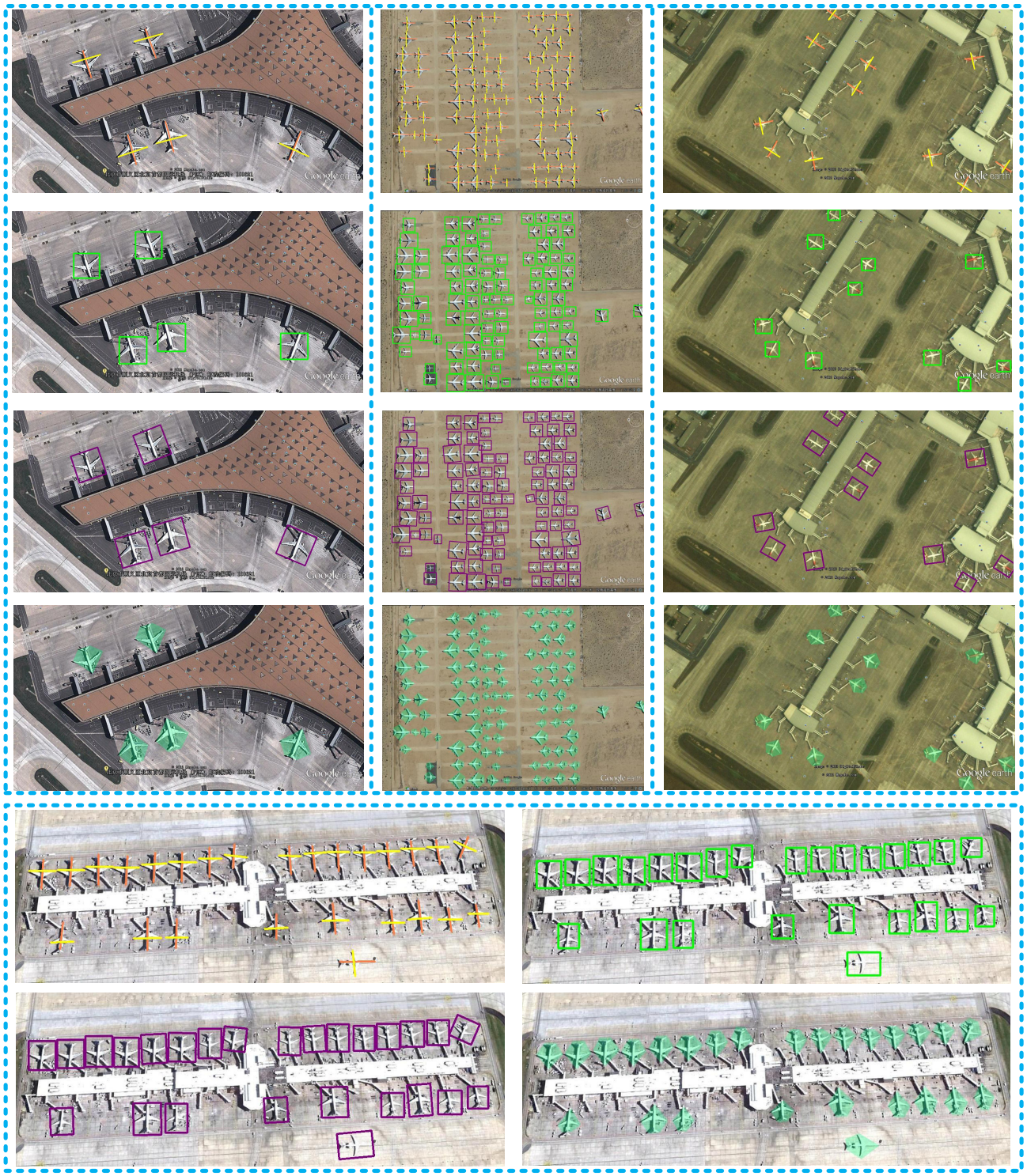


Fig. 13. Qualitative results output by X-LineNet.

4) Number of Prediction Modules: As shown in Fig. 3, each detection head is composed of three cascading prediction modules. In fact, for each detection head, the number of prediction modules can be adjusted flexibly. We use AP and frame per second (FPS) to analyze the influence of the number of prediction modules for X-LineNet.

As shown in Table V, the AP reaches the highest value of 95.1% when the cascading number is 5 or 8. However, the number of prediction modules affects the inference speed of X-LineNet. Hence, we choose the number of 3 in our experiments for the tradeoff of efficiency and accuracy.

5) Weight of Endpoints Detection Head Loss: The working of the Cutting-line Algorithm needs to predict four endpoints of line segments. When we run this algorithm, the total loss of X-LineNet is the accumulation of a weighted sum of the

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TABLE V
COMPARISONS OF DIFFERENT PREDICTION MODULES

number	1	3	5	8	10
AP	0.936	0.948	0.951	0.951	0.947
FPS	7.31	7.03	6.67	6.25	5.96

TABLE VI
COMPARISONS OF DIFFERENT WEIGHTS

λ_{cut}	0.1	0.3	0.5	1	2
AP	0.939	0.945	0.948	0.946	0.913

TABLE VII
COMPARISONS OF DIFFERENT WEIGHTS

λ_{pm}	0.1	0.3	0.5	1	2
Mask AP	0.869	0.893	0.890	0.874	0.857
AP	0.948	0.948	0.943	0.922	0.896

TABLE VIII
GENERATE THE PENTAGONAL MASK WITHOUT THE CUTTING-LINE ALGORITHM

λ_{pm}	0.1	0.3	0.5	1	2
Mask AP	0.853	0.862	0.869	0.855	0.848
AP	0.938	0.938	0.938	0.921	0.895

loss functions of line segments $\mathcal{L}_{ls}$ and endpoints $\mathcal{L}_{ep}$:

$$\mathcal{L}\text{oss} = \mathcal{L}_{ls} + \lambda_{\text{cut}}\mathcal{L}_{ep}. \tag{6}$$

We conduct an experiment to analyze the sensitivity of X-LineNet to the weight λ_{cut}.

Table VI shows the λ_{cut} of 0.5 yields the best 94.8% in AP. For larger values, the AP degrades, because they influence the convergence of the line segment's loss ($\mathcal{L}_{ls}$). Besides, too low weight (e.g., 0.1) affects the performance of the Cutting-line Algorithm and also yields an AP drop.

6) Loss Weight of Generating Pentagonal Mask: When we need to output the pentagonal mask, X-LineNet requires an additional detection head in parallel to predict the aircraft head position, as mentioned in Section: III-E. At this point, the total loss of X-LineNet is defined as

$$\mathcal{L}\text{oss} = \mathcal{L}_{ls} + \lambda_{\text{cut}}\mathcal{L}_{ep} + \lambda_{\text{pm}}\mathcal{L}_{\text{pm}} \tag{7}$$

where $\mathcal{L}_{\text{pm}}$ denotes the loss function of the pentagonal mask detection head, and λ_{pm} is the weight of it. We fix the λ_{cut} to 0.5 and analyze the sensitivity of X-LineNet to the weight λ_{pm}. Table VII shows that λ_{pm} of 0.3 can achieve the best 89.3% in Mask AP, and it does not affect the output accuracy (94.8% in AP) of the horizontal bounding box. Although low weight (e.g., 0.1) also has no impact on the AP, the Mask AP of pentagonal mask drops 2.4%. It is because the prediction of aircraft head is not accurate enough. For larger values, both the AP and Mask AP decrease due to a large λ_{pm} value affects the accuracy of line segments detection.

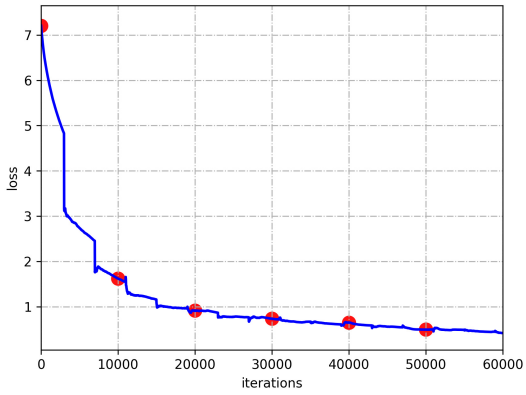


Fig. 14. Training loss of X-LineNet.

7) Generate the Pentagonal Mask Without the Cutting-Line Algorithm: The process of generating the pentagonal mask does not rely on the Cutting-line Algorithm. Therefore, we block the detection head of the Cutting-line Algorithm to only test the impact of the pentagonal mask one. In this case, the loss function of X-LineNet becomes

$$\mathcal{L}\text{oss} = \mathcal{L}_{ls} + \lambda_{\text{pm}}\mathcal{L}_{\text{pm}}. \tag{8}$$

According to the results in Tables VIII and II, for a low λ_{pm}, AP can still reach 93.8%. However, for large values, generating the pentagonal mask has a marked impact on the AP. Through the comparison of Tables VII and VIII, without the Cutting-line Algorithm, X-LineNet observes a Mask AP drop of 2.4% and an AP drop of 1%. Thus, the Cutting-line Algorithm is necessary.

8) Training Schedule: By default, we train X-LineNet for 60 000 iterations with a learning rate drop at 40 000 iterations. As shown in Fig. 14, these settings give an excellent convergence to the model. If we double the training iterations before dropping the learning rate, the performance only increases by 0.2% in AP, at the cost of much longer training time. Hence, we use the default number of iterations in our experiments to save computational resources.

9) NMS-Free: X-LineNet does not use complex IOU-based NMS in both training and test stages. To verify the model does not need NMS, we run NMS as a postprocessing step on the predictions, and the AP stays at 94.8%. Given no impact, our model does not use NMS. Incidentally, although the computational complexities of the clustering in X-LineNet and the NMS in other anchor-based detectors are both $O(n^2)$, the number of n is different. The n in X-LineNet is the number of objects, but in other anchor-based detectors, it also includes numerous redundant boxes. Thereby, X-LineNet is more efficient.

10) Low-Resolution Backbone: As mentioned in Section: IV-B, the output stride of all backbones (104-Hourglass, ResNet-101, and DLA-34) of X-LineNet is 4. We replace the backbone with a 32× ResNet-101 to verify the necessity of a high-resolution backbone for X-LineNet. Unfortunately, it yields an AP drop of 9% compared with the 4× ResNet-101, and thus a high-resolution backbone is essential.

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TABLE IX

STATE-OF-THE-ART COMPARISONS. AP^1 MEANS THE AP ACHIEVED IN TEST SETS OF AIRCRAFT-KP OR UCAS-AOD. AP^2 AND AP^3 CORRESPOND TO NWPU VHR-10, AND DIOR, RESPECTIVELY. AP^m IS THE AVERAGE OF AP^1, AP^2, AND AP^3. FPS MEANS FRAME PER SECOND

Models	Backbone	FPS	AP^1	AP^2	AP^3	AP^m	AP^1	AP^2	AP^3	AP^m
Horizontal Bounding Box			**Training on Aircraft-KP**				**Training on UCAS-AOD**			
Two-stage detectors										
Faster R-CNN [11]	vgg16	18	85.9%	85.6%	23.1%	64.9%	86.4%	85.2%	22.9%	64.8%
Faster R-CNN [15]	ResNet-50	15	88.7%	88.3%	34.3%	70.4%	88.4%	88.2%	34.7%	70.4%
Faster R-CNN [15]	ResNet-101	11	90.1%	90.5%	37.2%	72.6%	89.8%	90.4%	38.6%	72.9%
Faster R-CNN + FPN [22]	ResNet-101	7.1	91.3%	91.2%	41.1%	74.5%	92.4%	91.5%	40.9%	74.9%
Cascade R-CNN [49]	ResNet-101	3.7	92.8%	92.1%	46.6%	77.2%	93.1%	91.8%	44.1%	76.3%
One-stage detectors										
YOLO9000 [20]	DarkNet-19	21	82.2%	81.9%	38.9%	67.7%	82.4%	81.4%	39.6%	67.8%
YOLOv3 [21]	DarkNet-53	25	86.4%	86.2%	42.2%	71.6%	85.4%	85.5%	42.0%	71.0%
SSD [12]	ResNet-101	17	89.6%	89.4%	40.8%	73.3%	88.7%	89.4%	40.8%	73.0%
RetinaNe + FPN [13]	ResNet-101	7.2	90.1%	90.3%	41.7%	74.0%	89.8%	90.9%	41.2%	74.0%
ConerNet [19]	104-Hourglass	6.9	76.5%	75.4%	41.2%	64.4%	76.2%	77.4%	41.7%	65.1%
ExtremeNet [16]	104-Hourglass	5.4	78.3%	77.6%	37.5%	64.5%	78.3%	77.2%	37.2%	65.7%
X-LineNet	104-Hourglass	7.0	94.8%	94.1%	46.8%	78.6%	92.4%	91.9%	43.7%	76.0%
X-LineNet	ResNet-101	14	91.6%	91.2%	41.5%	74.8%	91.3%	91.1%	40.9%	74.4%
X-LineNet	DLA-34	27	92.5%	92.2%	42.4%	75.7%	91.9%	91.4%	41.8%	75.0%
Oriented Bounding Box			**Training on *Aircraft-KP***				**Training on *UCAS-AOD***			
RRPN [50]	ResNet-50	5.3	90.1%	-	-	-	90.4%	-	-	-
R2CNN [51]	ResNet-101	3.8	90.7%	-	-	-	90.9%	-	-	-
R-DFPN [52]	ResNet-101	4.6	91.0%	-	-	-	90.6%	-	-	-
X-LineNet	104-Hourglass	7.0	93.8%	-	-	-	92.1%	-	-	-
X-LineNet	ResNet-101	14	90.4%	-	-	-	90.2%	-	-	-
X-LineNet	DLA-34	27	91.2%	-	-	-	90.9%	-	-	-

F. Comparisons With State-of-the-Art Detectors

In this part, we compare X-LineNet with other state-of-the-art detectors. All data sets (mentioned in Section: IV-A) are utilized to prove the advancement of our model. We divide the experiment into two groups. One group uses the training set of Aircraft-KP to train the model, and the test sets include the Aircraft-KP, NWPU VHR-10, and DIOR. The other group trains the model with the UCAS-AOD, and its test sets include UCAS-AOD, NWPU VHR-10, and DIOR. We introduce four detailed definitions of AP. For these two groups of experiments, AP^1 denotes the AP achieved in test sets of Aircraft-KP or UCAS-AOD. AP^2 and AP^3 correspond to NWPU VHR-10 and DIOR, respectively. AP^m is the average of AP^1, AP^2, and AP^3. We experiment with all the backbone networks mentioned in Section: IV-B. X-LineNet does not need to output the pentagonal mask in this section, and thus we only keep three detection heads of it: two for detecting line segments and one for running the Cutting-line Algorithm. We set the parameters of X-LineNet according to the ablation studies (Section: IV-E). The sizes of X-LineNet under different backbone networks are 745M for 104-hourglass, 207M for ResNet-101, and 98M for DLA-34.

1) Training on Aircraft-KP: As shown in Table IX, X-LineNet with 104-hourglass achieves a 78.6% AP^m at a relatively good speed with 7.0 FPS, outperforming all reported anchor-based detectors and surpassing other two anchor-free detectors (CornerNet and ExtremeNet) by a large margin. X-LineNet with DLA-34 gives the best speed-accuracy trade-off. It runs at 27 FPS with 75.7% AP^m, which is 2 FPS faster than YOLOv3 and 4.1% AP^m more accurate. With ResNet-101, X-LineNet is about twice as fast at the same accuracy compared with RetinaNet.

For the output form of the oriented bounding box, we compare X-LineNet with several state-of-the-art oriented detectors. Our model with 104-Hourglass gives an AP^1 improvement of $\sim$ 3% compared with other rotated-anchor-based detectors. It is worth noting that the inference speed of X-LineNet is far faster than that of others. It is because other detectors use a more complex rotated-IOU calculation in the inference.

X-LineNet is a purely appearance-based line segments estimation network, without any implicit features learning, such as the height, width, and angle of the bounding box. This design enhances the learning of the visual grammar information of aircraft. Our model can still detect the target that is severely shielded or at the edge of the image, as shown in Fig. 15.

2) Training on UCAS-AOD: In order to prove the excellent performance of X-LineNet does not rely on the additional annotations in Aircraft-KP, we compare X-LineNet with other detectors via training on UCAS-AOD. As shown in Table IX, X-LineNet with 104-hourglass can still achieve

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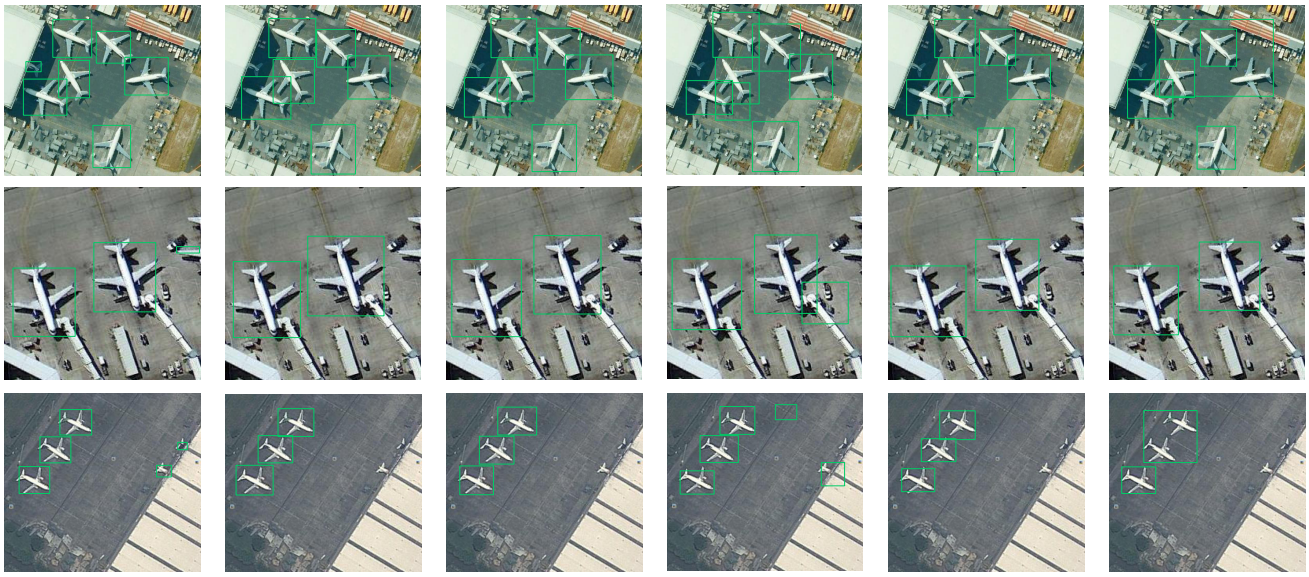


Fig. 15. X-LineNet can still detect the aircraft on an occasion that the target is severely shielded. (a) The effect of X-LineNet. (b), (c), (d), (e), and (f) Effects of Faster R-CNN, Cascade R-CNN, YOLOv3, RetinaNet, and CornerNet, respectively.

a good AP^m of 76.0%. Although the UCAS-AOD yields an AP^m drop of 2.6% compared with that of the Aircraft-KP data set, the accuracy still ahead of other one-stage detectors. In two-stage detectors, Cascade R-CNN is more accurate than X-LineNet by 0.3% in AP^m, but it is also twice as slow. This experiment proves that the robustness of X-LineNet on different training sets is satisfactory.

V. Conclusion

In conclusion, we present a novel one-stage and anchor-free model named X-LineNet, which detects aircraft in a bottom-up approach. The task of aircraft detection is transformed into the prediction and clustering of a pair of appearance-based intersecting line segments by X-LineNet. This design enables the model to learn rich and specific visual grammar information of the target. As a result, our model outperforms state-of-the-art one-stage detectors and is still competitive compared with advanced two-stage detectors in the field of aircraft detection. In the end, we encourage the researchers to rethink the necessity of anchor-based RPN frameworks in RS object detection, which is currently considered as the de facto standard for detection in this field.

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